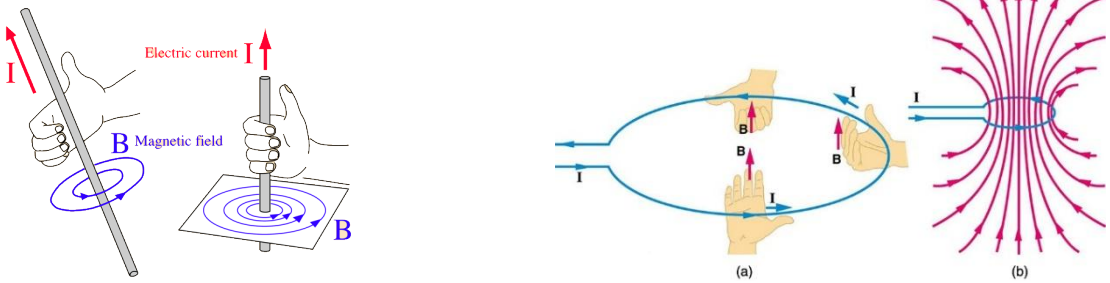
Chapter 47 - Current and the Magnetic Field
Calculus Physics II

Magnetic Fields Produced by Current - Magnetic Fields are produced by moving charges or currents. When a current flows in a wire the moving charge produces a magnetic field around that wire. The shape of the magnetic field depends on the configuration of the wire and the direction of the current.

The direction of the magnetic field for a current carrying wire is determined by the 2nd right hand rule.

Right hand rule #2 (RHR # 2)– Point your thumb in the direction of the positive current flow and your fingers wrap around the wire, the tips of your fingers point in the direction of the magnetic field.



For the straight wire above, the current direction is up so point the thumb in this direction and wrap the fingers around the wire, this is the circular direction of the $\vec{B}$-field.

For the loop of wire the current is moving counterclockwise, pick any point on the wire and point the thumb in the direction of the current, wrap your fingers around the wire, this is the circular direction of the $\vec{B}$-field in the center of the loop of wire.

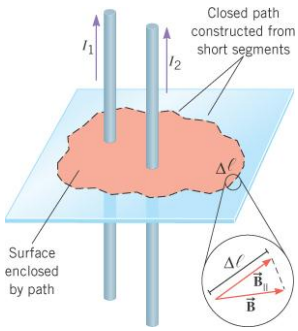
- The RHR # 2 is used to determine the direction of the magnetic field from a current carrying wire or the direction of the current when given the magnetic field direction.

Ampere's Law for Magnetic Fields - Ampere's Law is a fundamental principle in electromagnetism that relates the circulation of a magnetic field around a closed loop to the electric current passing through that loop. It is one of Maxwell's equations and is important for understanding how magnetic fields are generated by electric currents.

$$\oint \vec{B} \bullet d\vec{L} = \mu_o I_{enclosed}$$

where $d\vec{L}$ is a small segment of length along a closed path of arbitrary shape around the current, B_{ll} is the component of the magnetic field parallel to $d\vec{L}$ at the Amperian loop. I is the net current passing through the surface bounded by the path and μ_o is the permeability of free space.

- In making the choice of this loop, the $\vec{B}$ field must be parallel to the direction of $d\vec{L}$ (tangent to the loop), when this condition is met then the integral simplifies to the magnitude of the magnetic field times the circumference of the Amperian loop.
- Only the current passing through the area enclosed by the Amperian loop contributes to the integral.
- Currents outside the loop do not contribute to the $\vec{B}$-field.
- The Magnetic field direction at the loop and dL is determined by the right hand rule # 2 using the current direction.



Example: For the current carrying wires above, given that $I_1 = 2$ A and $I_2 = 3$ A,

A) Determine the value for Ampere's integral for the surface shown.

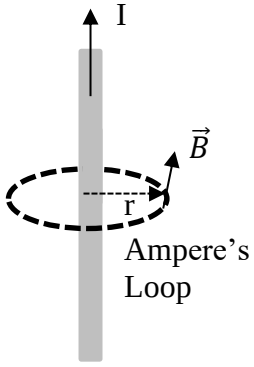
$$\oint \vec{B} \bullet d\vec{L} = \mu_o I_{enclosed} = (1.26\ x\ 10^{-6}\ H/m)(2\ A + 3\ A) = 6.3\ x\ 10^{-6}\ HA/m$$

B) Now assume that the current in wire 1 is in the opposite direction, determine the value for Ampere's integral.

$$\oint \vec{B} \bullet d\vec{L} = \mu_o I_{enclosed} = (1.26\ x\ 10^{-6}\ H/m)(3\ A - 2\ A) = 1.26\ x\ 10^{-6}\ HA/m$$

Example; Using the symmetry of a long straight wire a Amperian loop can be drawn at a radius r from the wire to determine the magnetic field strength at that location. For a long straight wire carrying current I, determine the magnetic field strength at a distance r from the wire.

Ampere's Loop is drawn a distance r from the center of the wire. The circumference of the circle is C = L = 2πr



The magnetic field direction from RHR # 2 is into the page and perpendicular to r. dL is in the same direction as B therefore the dot product is

$$\oint \vec{B} \bullet d\vec{L} = \oint |\vec{B}| \cos(0°) dL = |\vec{B}|(2\pi r) = \mu_o I_{enclosed}$$

Solving for the magnetic field strength for a long straight current carrying wire

$$|\vec{B}| = \frac{\mu_o I}{2\pi r}$$

Biot-Savart Law (Pronounced "Bee-oh suh var") describes the magnetic field generated by a constant electric current considering the contribution from each small segment of the current-carrying conductor ($d\vec{L}$). The law states that the magnetic field ($d\vec{B}$) from a current element ($Id\vec{l}$) at a point is proportional to the current, the length of the element, the sine of the angle between the current and the position vector, and inversely proportional to the square of the distance from the element. The direction of the magnetic field is given by the cross product of the current element vector and the unit position vector $d\vec{L}x\hat{r}$, or by using the 2nd right-hand rule.

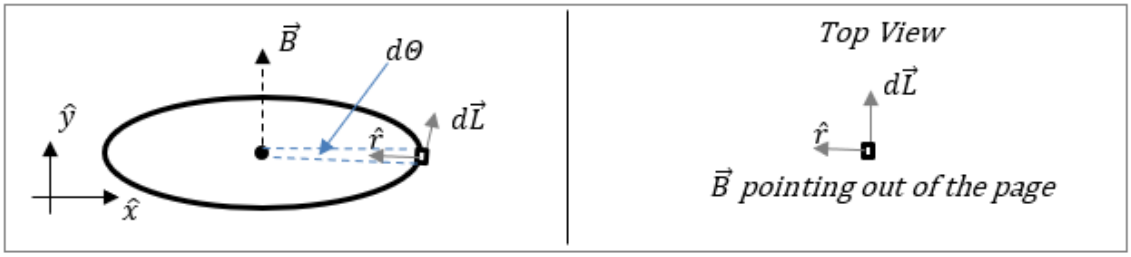
The Vector Form for the magnetic field due to a current carrying wire:

$$d\vec{B} = \left[\frac{\mu_o}{4\pi}\right]\frac{I\left(d\vec{L}x\hat{r}\right)}{r^2} \qquad \vec{B} = \frac{\mu_o}{4\pi}\int\frac{I\left(d\vec{L}x\hat{r}\right)}{r^2}$$

$d\vec{B}$ is the infinitesimally small magnetic field produced by a current element.
μ_o is the magnetic permeability of free space, a constant equal to 1.26×10^{-6} T·m/A.
I is the electric current flowing through the wire.
$d\vec{L}$ is an infinitesimally small vector representing the length and direction of the current element.
$\hat{r}$ is the unit vector pointing from the current element to the point where the magnetic field is being calculated.
r is the distance from the current element to the point in space were the magnetic field is going to the calculated.

- Instead of treating the entire current-carrying wire, the Biot-Savart law breaks it down into infinitesimal segments $Id\vec{L}$.
- Similar to Coulomb's Law and Newton's Law of Gravitation, the magnetic field strength decreases with the square of the distance from the current element.
- To find the direction of the magnetic field $d\vec{B}$, the direction of the cross product of $d\vec{L}x\hat{r}$ or use your right-hand rule # 2.
- To find the total magnetic field from a conductor of finite size or shape, you must integrate this expression for $d\vec{B}$ over the length of the wire.

Example: The magnetic field at the center of a circular current carrying wire, the current is flowing counterclockwise.



Determine the magnetic field strength at the center of a circular current carrying wire with radius of R.

$$\vec{B} = \frac{\mu_o}{4\pi}\int \frac{I\left(d\vec{L}x\hat{r}\right)}{r^2}$$

$$d\vec{L}x\hat{r} = \left[\left|d\vec{L}\right|\sin(\varphi)\right]\hat{y}$$

Were the angle φ is between the infinitesimal length vector $d\vec{L}$ and the unit radius vector $\hat{r}$. From the "Top View" in the diagram above the angle between he vectors is equal to 90°, which makes the sin(φ) = 1. The infinitesimal angle that spans dL is given by

$$d\Theta = \frac{dL}{R} \quad \text{Therefore } dL = Rd\Theta$$

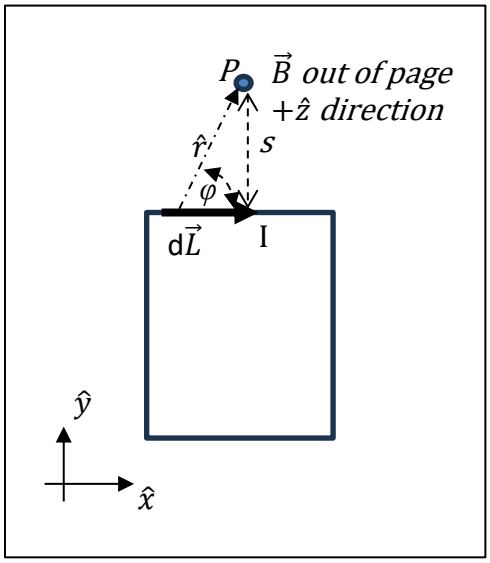
Using the RHR # 2 the magnetic field at the center of the wire loop will be perpendicular to both $d\vec{L}$ and $\hat{r}$. The magnetic field direction will be perpendicular to the surface area of the loop. This will be defined as the positive y direction.

$$B_y = \frac{\mu_o}{4\pi}\int_0^{2\pi} \frac{I(Rd\Theta)}{R^2}Rd\Theta = \frac{I\mu_o}{4\pi R}\int_0^{2\pi} d\Theta$$

For the magnetic field strength at the center of a circular current carrying wire

$$B_y = \frac{2\pi RI\mu_o}{4\pi R^2} = \frac{I\mu_o}{2R}$$

Example: A rectangular loop of wire has a current I flowing clock wises through it, determine the magnetic field strength a distance "s" from the top of the wire with length L. Ignore the magnetic field from the side and bottom wires.



Determining the incremental length of the wire.

$$d\vec{L} = d\vec{x}$$

The cross product gives a magnitude along the positive z axis

$$d\vec{L}x\hat{r} = d\vec{x}x\hat{r} = [dxsin(\varphi)]\hat{z}$$

where the angle φ is between the incremental length d$\vec{L}$ and the radius unit vector $\hat{r}$. Putting the radius r and angle φ in terms of the integration variable x

$$r^2 = (x^2 + s^2)$$

$$\sin(\varphi) = \frac{h_o}{h} = \frac{s}{r} = \frac{s}{(x^2 + s^2)^{1/2}}$$

To determine the magnetic field strength using the Biot-Savart Law

$$\vec{B} = \frac{\mu_o I}{4\pi} \int \frac{\left(d\vec{L} x \hat{r}\right)}{r^2}$$

Substituting for the radius and angle

$$B_z = \frac{\mu_o I}{4\pi} \int \frac{[dx sin(\varphi)]\hat{z}}{r^2} = \frac{\mu_o I}{4\pi} \int \left[\frac{1}{(x^2 + s^2)}\right]\left[\frac{s}{(x^2 + s^2)^{1/2}}\right] dx$$

Taking the integral from -L/2 to +L/2

$$B_z = \frac{\mu_o I}{4\pi} \int_{-L/2}^{l/2} \left[\frac{s}{(x^2 + s^2)^{3/2}}\right] dx$$

Using the equation for the integral

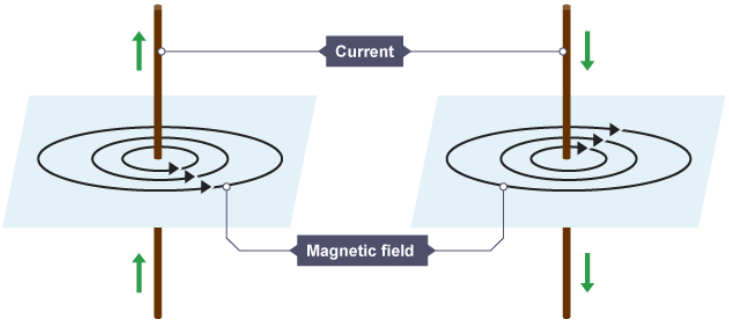
$$\int \frac{1}{(x^2 + a^2)^{3/2}} dx = \frac{x}{a^2(x^2 + a^2)^{1/2}}$$

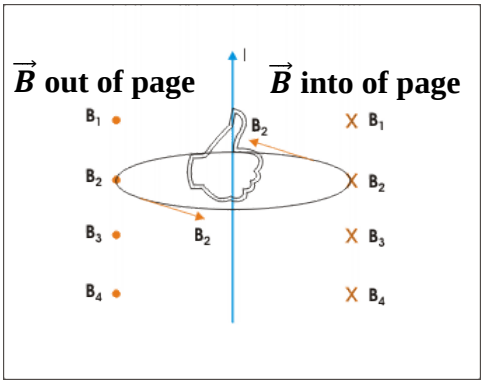
Solving and simplifying the equation

$$B_z = \frac{\mu_o I s}{4\pi} \left[\frac{x}{s^2(x^2 + s^2)^{1/2}}\right]_{-L/2}^{L/2} = \frac{\mu_o I}{4\pi s} \left[\frac{L}{((L/2)^2 + s^2)^{1/2}}\right]$$

Magnetic field strengths for common wire configurations - Magnetic field strengths produced by electric currents depend on the wire's geometry, the amount of current (I),the distance (r) from the conductor and the number of loops of wire (N) that are used.

Magnetic Field produced by a straight wire- When current flows through a straight wire the magnetic field that is produced is concentric circles moving out from the center of the wire and is weaker the father from the wire. For the view of a wire in the plane of the page, if the current is moving towards the top of the page then the field to the right of the wire is into the page and the field to the left of the wire is coming out of the page.





<u>Magnitude of the magnetic field around a long straight wire</u> depends on the current in the wire and the distance from the wire derived from Ampere's Law.

$$|\vec{B}| = \frac{\mu_o I}{2\pi r}$$

μ_o = permeability of free space for vacuum/air
$= 4\pi \times 10^{-7}$ Tm/A
$\mu_o = 1.257 \times 10^{-6}$ Tm/A
r = distance to point (meters)
I = current flowing in wire (Amps)

<u>Example</u>: A long straight wire has a current of 2 A flowing upward as shown,



What is the magnitude and direction of the magnetic field 1 cm right of the wire?

$$B = \frac{\mu_o I}{2\pi r} = \frac{(1.257 \times 10^{-6}\ \text{Tm/A})(2\ \text{A})}{2\pi(0.01\ \text{m})} = 4\ x\ 10^{-5}\ T\ into\ page$$

What is the magnitude and direction of the magnetic field 1 m left of the wire?

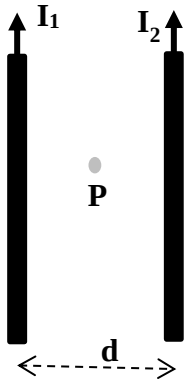
$$B = \frac{\mu_o I}{2\pi r} = \frac{(1.257 \times 10^{-6}\ \text{Tm/A})(2\ \text{A})}{2\pi(1\ \text{m})} = 4\ x\ 10^{-7}\ T\ out\ of\ page$$

<u>Net magnetic Field due to multiple wires</u> – The vector sum of each wires magnetic field at that point in space due to current sources.

$$\Sigma\vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 + \cdots$$

<u>Example</u>: Two long straight wires are spaced d = 20 cm apart, if wire 1 has I_1 = 2 A flowing upward and wire 2 has I_2 = 5 A flowing upward, what is the net magnetic field at point P halfway between the two wires?

Calculate the magnitude and direction of the $\vec{B}$ field due to wire 1 at point P.



$$|\vec{\mathbf{B}}_1| = \frac{\mu_o I_1}{2\pi r_1} = \frac{(1.257 \times 10^{-6}\ \mathrm{Tm/A})(2\ \mathrm{A})}{2\pi(0.01\ \mathrm{m})} = 4\ x\ 10^{-5}\ T\ into\ page$$

Calculate the magnitude and direction of the $\vec{B}$ field due to wire 2 at point P.

$$|\vec{\mathbf{B}}_2| = \frac{\mu_o I_2}{2\pi r_2} = \frac{(1.257 \times 10^{-6}\ \mathrm{Tm/A})(5\ \mathrm{A})}{2\pi(0.01\ \mathrm{m})} = 1\ x\ 10^{-4}\ T\ out\ of\ page$$

$$\Sigma\vec{\mathbf{B}} = \vec{\mathbf{B}}_1 + \vec{\mathbf{B}}_2 + \vec{\mathbf{B}}_3 + \cdots$$

In the z direction (in or out of page) and assuming out of the page is positive then

$$\Sigma B_z = (-B_1) + (+B_2) = -4x10^{-5}T + 1x10^{-4}T = +6x10^{-5}T\ out\ of\ page$$

<u>Forces between two wires</u> - For two long straight wires that are close and parallel: the force between them is attractive if the current is in the same direction and repulsive if the current is in the opposite direction. This attractive and repulsive force is due to the interaction of the current in the first wire with the magnetic field produced by the second wire. For the second wire it is the current in the second wire interacting with the magnetic field produced by the first wire.

The magnetic field created by each wire at the location of the other is given by

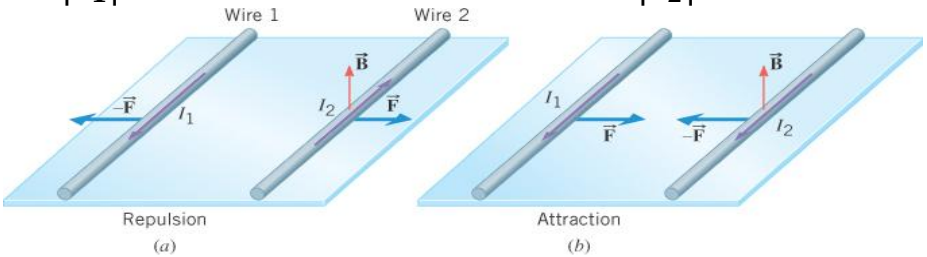
$$|\vec{\mathbf{B}}_1| = \frac{\mu_o I_1}{2\pi r} \qquad |\vec{\mathbf{B}}_2| = \frac{\mu_o I_2}{2\pi r}$$

Where r is the distance between the wires

<u>The direction</u> of the magnetic field created by the current of the first wire is given by the RHR # 2 using the current direction.

The force each wire experience due to the field created by the other wire is

$|\vec{\mathbf{F}}_1| = I_1 L B_2$ $\qquad$ $|\vec{\mathbf{F}}_2| = I_2 L B_1$



- ✓ B_1 is the magnetic field created by wire # 1
- ✓ B_2 is the magnetic field created by wire # 2

- ✓ F_1 is the force on wire one from the current in wire one by the magnetic field created by wire 2
- ✓ F_2 is the force on wire two from the current in wire two by the magnetic field created by wire 1

The direction of the force on the current carrying wire is given by the RHR # 1 using the current in that wire and the magnetic field produced by the other wire.

Example: If wire one is L_1 = 2 m long and has a current of I_1 = 1 A flowing through it, wire one is r = 1 cm from wire 2 which has I_2 = 2 A of current and is also L_2 = 2 m long, what is the force on each wire due to the other.

$$B_1 = \frac{\mu_o I_1}{2\pi r} = \frac{(1.257 \text{ x } 10^{-6} \text{ Tm/A})(1 \text{ A})}{2\pi(0.01 \text{ m})} = 2\ x\ 10^{-5}\ T\ at\ wire\ 2$$

$$B_2 = \frac{\mu_o I_2}{2\pi r} = \frac{(1.257 \text{ x } 10^{-6} \text{ Tm/A})(2 \text{ A})}{2\pi(0.01 \text{ m})} = 4\ x\ 10^{-5}\ T\ at\ wire\ 1$$

$F_1 = I_1LB_2 = (1A)(2m)(4 \text{ x } 10^{-5} \text{ T}) = 8 \text{ x } 10^{-5} \text{ N}$

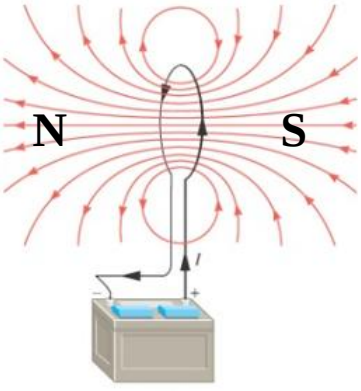
$F_2 = I_2LB_1 = (2A)(2m)(2 \text{ x } 10^{-5} \text{ T}) = 8 \text{ x } 10^{-5} \text{ N}$

The direction of the force depends on the direction of the currents

Currents in the opposite direction are repulsive.
Currents in the same direction are attractive.

Center of circular coil or loop or wire - For a wire bent in the shape of a circle with radius R and N loops of wire around the circle, the magnetic field at the center of the loop is given by

$$|\vec{B}| = N\left(\frac{\mu_o I}{2R}\right)$$



$|\vec{B}|$ magnitude of the magnetic field at the center of the loop.
R is the radius of the loop in meters
N is the number of turns of wire or loops or wire
I is the current through the wire
$\mu_o = 1.257 \text{ x } 10^{-6}$ Tm/A

- The magnetic field of the current loop is very similar to the bar magnet and the direction is determined by the RHR #2 using the current direction!

- The magnetic field lines outside the loop go from what acts as the north pole to what acts as the south pole.

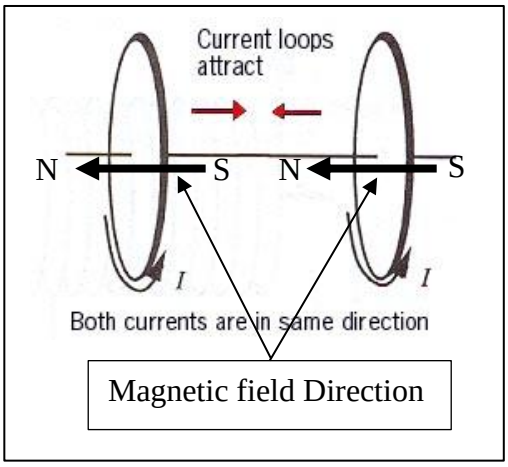
Example: If a coil of wire with a radius of 0.01 m has 10 turns and 2 A flowing through it, what is the magnitude of the magnetic field at its center?

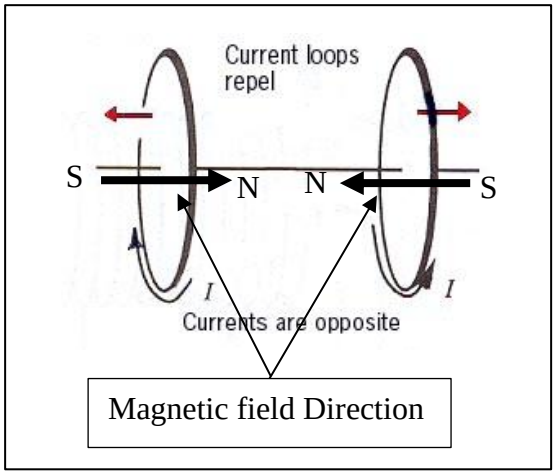
$$\mathbf{B} = \mathbf{N}\left(\frac{\mu_o\, I}{2R}\right) = \frac{(1.257 \times 10^{-6}\ \mathrm{Tm/A})(2\ \mathrm{A})(10)}{2(0.01\ \mathrm{m})} = 1.26\ x\ 10^{-3}\ T$$

Example: When two current loops are brought near depending on the current direction, they will either attract or repel just like a bar magnet.

The First two loops the current is in the same direction therefore the magnetic field (right hand rule # 2) is in the same direction for both, the "North" would be to the left of the first loop and the "South" would be to the right. This would be the same for the second loop therefore the "south" of the first loop is on the same side as the "North" of the second loop and they Attract!

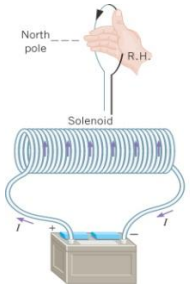
The Second two loops the current is in the opposite directions therefore the magnetic field (right hand rule # 2) is in the opposite direction and the "North" would be to the right of the first loop and the "South" would be to the left. This would be the opposite for the second loop therefore the "North" of the first loop is on the same side as the "North" of the second loop and they Repel!

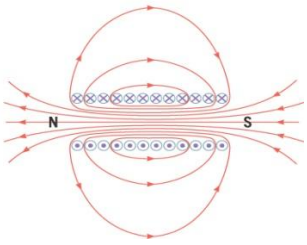




Interior point of long solenoid – A solenoid is a long tube that has wire wrapped around. When a current is passed through the wire of the solenoid it produces inside the tube a very uniform magnetic field. For a solenoid with N loops of wire the magnetic field inside is given by

$$|\vec{B}| = N\left(\frac{\mu_o\, I}{L}\right)$$





- the magnetic field $\vec{\mathbf{B}}$ is constant in the interior, it is the same though out the inside of the tube

- the magnetic field $\vec{\mathbf{B}}$ is nearly zero outside the solenoid

Sometimes the solenoid equation is given with a loop density "n"

$$|\vec{B}| = \mu_o nI$$

Where n is the number of turns per length of the solenoid n = N/L

Example: If a 0.1 m long solenoid filled with air has a current of 2 A with 1,000 loops what is the magnitude of the magnetic field inside the solenoid?

$$B = \frac{\mu_o\, IN}{L} = \frac{(1.257 \text{ x } 10^{-6}\ \text{Tm/A})(2\ \text{A})(1000)}{(0.1\ \text{m})} = 0.025\ T$$

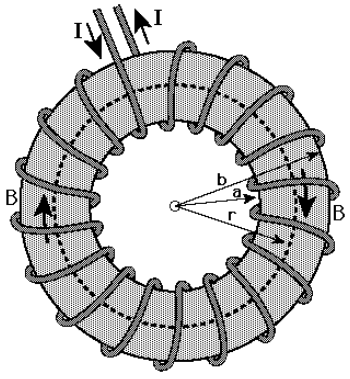
What if instead of the solenoid being filed with air it is filed with a iron cylinder, iron has a magnetic permittivity of $\mu_{iron} = 6.5 \text{ x } 10^{-3}$ Tm/A

$$B = \frac{\mu_{iron}\, IN}{L} = \frac{(6.5 \text{ x } 10^{-3}\ \text{Tm/A})(2\ \text{A})(1000)}{(0.1\ \text{m})} = 130\ T$$

The addition of an iron ferromagnetic core increased the magnitude of the magnetic field by over 5,000 time!

Interior magnetic field of a Toroid – A toroid can be thought of as a solenoid that has been bent around until the ends connect. For a toroid having N loops of wire where r is the radius of the circle the toroid makes. The magnetic field inside is given by

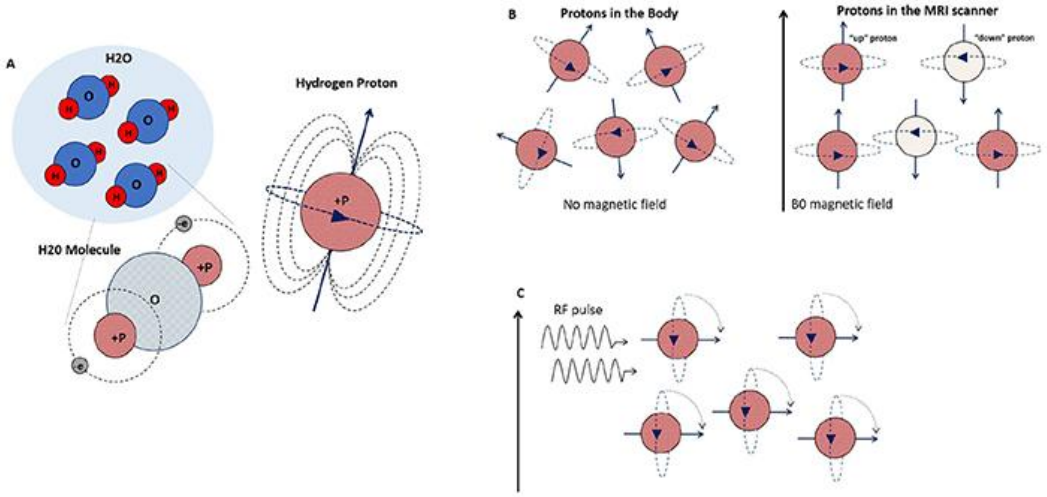
$$|\vec{B}| = N\left(\frac{\mu_o I}{2\pi r}\right)$$



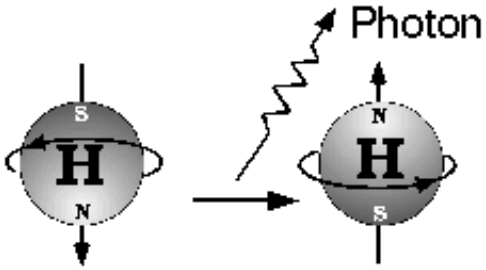
Where **r** is the radius of the toroid in meters
N is the number of turns of wire or loops or wire
I is the current through the wire
$\mu_o = 1.257 \text{ x } 10^{-6}$ Tm/A

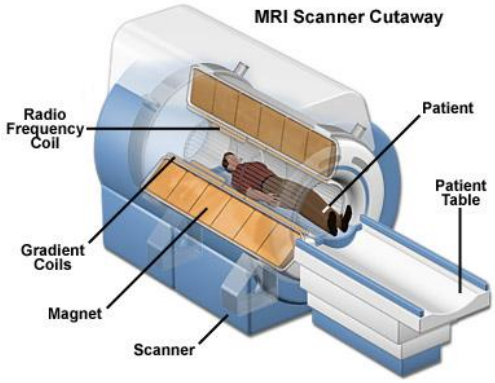
Applications for Magnetic fields

MRI - Magnetic Resonance Imaging - The MRI uses the hydrogen atoms in the body, each hydrogen has something known as spin, the hydrogen atoms in the body are exposed to a high external magnetic field this external field aligns the hydrogen atoms along the field lines due to the magnetic field created by the spin of the hydrogen atom. Some of the hydrogen atoms align in the up state (high energy state) and other on the down state (low energy state).

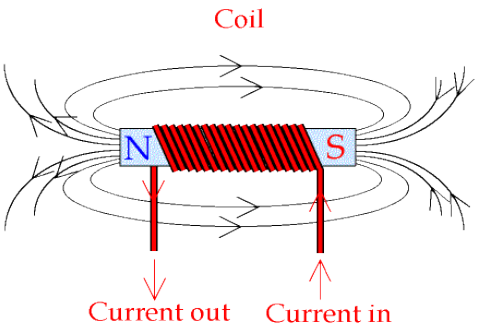


By sending a high frequency magnetic field (probe field or Resonance field) the down spin atoms (low energy) will be flipped to the up state (high energy) by absorbing the energy from the probe magnetic field. The probe beam is turned off, the atoms that were flipped to the high energy state will flip back to the down state, by doing this they release the energy they had absorbed as a high energy photon (gamma ray). This energy is detected and mapped out. The higher density material will have more hydrogen atoms therefore the more energy is emitted from these regions. The MRI maps the densities in the body over a very thin cross section.





Electromagnet – When current passes through a current loop it produces a magnetic field that looks very similar to a bar magnet. If the center of the loop is filled with a ferromagnetic material, then the magnetic field will be greatly increased due to the material. The more loops of wire are used the more magnetic field will be produced and the larger the electromagnet. Once the current is stopped the magnetic field will go away and there will not be a magnet.





<u>Tape recorder/credit card reader/ Hard drive on computer</u> - A tape recorder or credit card reader uses a magnetic material that is magnetizes to recording data from an electric signal. This magnetized data is read back by sensing the magnetic field from the recording.

